

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH `adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT fields	SoA bufs	End-to-end			Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.6896	0.6823	1.01×	0.3853	0.3284	1.17×	0.3042	0.3538	0.86×
Compiler	9	8	0.2970	0.2215	1.34×	0.1358	0.0401	3.39×	0.1612	0.1814	0.89×
DBQuery	15	13	0.2631	0.2542	1.04×	0.1258	0.0862	1.46×	0.1373	0.1680	0.82×
DecisionTree	7	6	3.941	2.955	1.33×	3.941	2.955	1.33×	—	—	—
DomTree	15	14	0.2069	0.1396	1.48×	0.1484	0.0618	2.40×	0.0585	0.0778	0.75×
KDTree	17	16	0.1890	0.1401	1.35×	0.1890	0.1401	1.35×	—	—	—
LinearListReduction	11	11	0.0319	0.0050	6.35×	0.0319	0.0050	6.35×	—	—	—
List	2	2	0.3875	0.4577	0.85×	0.1333	0.1153	1.16×	0.2542	0.3424	0.74×
MonoTree	3	2	0.0636	0.0661	0.96×	0.0178	0.0229	0.78×	0.0459	0.0432	1.06×
ObjectGraph	5	4	0.2403	0.2437	0.99×	0.0977	0.0820	1.19×	0.1427	0.1616	0.88×
PiecewiseFunctions	8	7	0.3860	0.3623	1.07×	0.1466	0.0951	1.54×	0.2393	0.2672	0.90×
TernaryTree	5	3	0.1121	0.1045	1.07×	0.0286	0.0261	1.09×	0.0835	0.0784	1.07×
Trie	10	9	0.2205	0.1994	1.11×	0.0679	0.0327	2.08×	0.1525	0.1667	0.91×
ColorOctree	25	18	0.0944	0.0769	1.23×	0.0944	0.0769	1.23×	—	—	—

Table 2: Native PAPI metrics speedup summary across all passes. For each program and metric, speedup is AoS/SoA where each side is the sum of per-pass median counter values. >1× means SoA reports fewer events.

Program	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
OctTree	1.05×	0.75×	1.34×	0.48×	0.26×	0.48×	3.41×
Compiler	1.64×	0.52×	4.01×	0.64×	15.21×	0.35×	6.73×
DBQuery	1.10×	0.47×	1.32×	0.34×	8.48×	0.21×	6.09×
DecisionTree	1.33×	0.59×	17.06×	3.20×	1.03×	3.34×	36.20×
DomTree	1.65×	0.57×	2.43×	1.00×	20.19×	0.29×	10.19×
KDTree	1.35×	0.77×	1.47×	0.84×	0.22×	0.85×	4.27×
LinearListReduction	6.38×	0.90×	35.11×	5.81×	35.74×	5.69×	11.38×
List	0.81×	0.80×	3.58×	0.47×	0.11×	0.70×	1.46×
MonoTree	0.97×	0.85×	0.50×	1.08×	0.27×	1.02×	1.21×
ObjectGraph	0.99×	0.75×	6.14×	0.77×	0.53×	0.58×	2.81×
PiecewiseFunctions	1.10×	0.60×	2.23×	1.48×	1.89×	0.30×	4.40×
TernaryTree	1.14×	0.79×	0.30×	0.48×	0.22×	0.89×	1.33×
Trie	1.17×	0.56×	3.95×	0.97×	2.98×	0.08×	6.11×
Geomean	1.33×	0.67×	2.72×	0.94×	1.43×	0.60×	4.63×

Table 3: Native PAPI metrics totals across all passes (baseline variants). Each entry is (Am/Sm), where Am and Sm are sums of per-pass median counter values.

Program	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
OctTree	2.7e+09/ 2.5e+09	8.2e+09 /1.1e+10	1.4e+07/ 1.1e+07	4.1e+05 /8.4e+05	6.6e+04 /2.6e+05	1.3e+04 /2.7e+04	6.4e+07/ 1.9e+07
Compiler	9.9e+08/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.3e+06	2.2e+05 /3.4e+05	1.3e+06/ 8.5e+04	1.4e+04 /4.1e+04	5.0e+07/ 7.5e+06
DBQuery	9.6e+08/ 8.8e+08	1.1e+09 /2.2e+09	3.9e+06/ 3.0e+06	1.5e+05 /4.5e+05	8.2e+05/ 9.7e+04	6.7e+03 /3.2e+04	2.1e+07/ 3.4e+06
DecisionTree	2.0e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.2e+07	1.4e+05/ 4.4e+04	2.2e+06/ 2.1e+06	1.4e+05/ 4.2e+04	1.1e+09/ 3.0e+07
DomTree	8.8e+08/ 5.3e+08	1.2e+09 /2.2e+09	1.3e+07/ 5.2e+06	1.1e+05/ 1.1e+05	1.1e+06/ 5.2e+04	8.6e+03 /2.9e+04	5.6e+07/ 5.5e+06
KDTree	9.6e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.4e+07/ 9.9e+06	9.4e+02 /1.1e+03	2.3e+04 /1.0e+05	9.5e+02 /1.1e+03	4.4e+07/ 1.0e+07
LinearListReduction	1.6e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.6e+06/ 4.6e+04	2.7e+02/ 4.7e+01	2.5e+05/ 7.1e+03	2.7e+02/ 4.8e+01	1.2e+07/ 1.1e+06
List	1.3e+09 /1.6e+09	4.5e+09 /5.6e+09	9.2e+06/ 2.6e+06	2.9e+05 /6.1e+05	2.2e+04 /2.0e+05	1.3e+04 /1.8e+04	4.9e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.2e+05 /4.5e+05	5.0e+04/ 4.6e+04	3.6e+03 /1.3e+04	2.1e+03/ 2.1e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.7e+08 /8.8e+08	3.0e+09 /4.0e+09	9.2e+06/ 1.5e+06	2.1e+05 /2.7e+05	2.9e+04 /5.4e+04	1.1e+04 /1.8e+04	2.4e+07/ 8.4e+06
PiecewiseFunctions	1.3e+09/ 1.2e+09	2.8e+09 /4.7e+09	1.4e+07/ 6.5e+06	5.6e+05/ 3.8e+05	1.8e+05/ 9.5e+04	1.3e+04 /4.2e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	4.8e+05 /1.6e+06	6.8e+04 /1.4e+05	4.4e+03 /2.0e+04	2.5e+03 /2.8e+03	3.3e+06/ 2.5e+06
Trie	6.7e+08/ 5.7e+08	1.2e+09 /2.2e+09	9.4e+06/ 2.4e+06	2.7e+05 /2.8e+05	1.5e+05/ 5.1e+04	3.9e+03 /4.9e+04	2.5e+07/ 4.2e+06

Table 4: Baseline Gibbon comparison (times only). Shown variants: AoS-mut, AoS-imm, SoA-mut. Times are full executable end-to-end wall time per run (s), not pass-sum time. **Bold** marks the fastest time across shown variants.

Program	Am	Ai	Sm
Compiler	31.08	114.2	23.97
DBQuery	27.35	34.29	27.20
DecisionTree	394.3	577.9	294.9
DomTree	21.52	35.59	14.83
KDTree	19.36	26.00	14.22
LinearListReduction	3.512	14.58	0.8810
List	40.54	<i>OOM</i>	48.44
MonoTree	6.877	14.65	7.165
ObjectGraph	24.97	49.40	25.81
Oct:sumMass	4.086	4.868	3.546
Oct:sumEnergy	4.302	5.993	4.095
Oct:countActive	4.069	5.284	3.809
Oct:countParticles	4.228	5.082	3.702
Oct:barnesHutPotential	4.642	6.244	4.948
Oct:fmmPotential	10.21	10.13	10.72
Oct:scaleEnergy	16.59	16.74	20.00
Oct:clearFlags	17.08	16.95	19.86
PiecewiseFunctions	40.34	57.91	38.32
TernaryTree	12.15	15.16	11.26
Trie	23.52	31.67	21.68
ColorOctree	9.807	12.65	8.440

Table 5: Baseline Gibbon comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut. Speedups are computed from full executable end-to-end wall time per run. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm
Compiler	1.30 $\times$	3.67 $\times$	4.76 $\times$
DBQuery	1.01 $\times$	1.25 $\times$	1.26 $\times$
DecisionTree	1.34 $\times$	1.47 $\times$	1.96 $\times$
DomTree	1.45 $\times$	1.65 $\times$	2.40 $\times$
KDTree	1.36 $\times$	1.34 $\times$	1.83 $\times$
LinearListReduction	3.99 $\times$	4.15 $\times$	16.55 $\times$
List	0.84 $\times$	–	–
MonoTree	0.96 $\times$	2.13 $\times$	2.04 $\times$
ObjectGraph	0.97 $\times$	1.98 $\times$	1.91 $\times$
Oct:sumMass	1.15 $\times$	1.19 $\times$	1.37 $\times$
Oct:sumEnergy	1.05 $\times$	1.39 $\times$	1.46 $\times$
Oct:countActive	1.07 $\times$	1.30 $\times$	1.39 $\times$
Oct:countParticles	1.14 $\times$	1.20 $\times$	1.37 $\times$
Oct:barnesHutPotential	0.94 $\times$	1.35 $\times$	1.26 $\times$
Oct:fmmPotential	0.95 $\times$	0.99 $\times$	0.95 $\times$
Oct:scaleEnergy	0.83 $\times$	1.01 $\times$	0.84 $\times$
Oct:clearFlags	0.86 $\times$	0.99 $\times$	0.85 $\times$
PiecewiseFunctions	1.05 $\times$	1.44 $\times$	1.51 $\times$
TernaryTree	1.08 $\times$	1.25 $\times$	1.35 $\times$
Trie	1.08 $\times$	1.35 $\times$	1.46 $\times$
ColorOctree	1.16 $\times$	1.29 $\times$	1.50 $\times$

Table 6: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
barnesHutPotential	F	11/16	31%	0.0428 $\pm$ 0.00001	0.0586 $\pm$ 0.00017	0.0394$\pm$0.00109	1.09 $\times$	1.37 $\times$	1.49 $\times$
countActive	F	10/16	38%	0.0369 $\pm$ 0.00014	0.0466 $\pm$ 0.00166	0.0291$\pm$0.00002	1.27 $\times$	1.26 $\times$	1.60 $\times$
countParticles	F	8/16	50%	0.0378 $\pm$ 0.00100	0.0461 $\pm$ 0.00032	0.0278$\pm$0.00002	1.36 $\times$	1.22 $\times$	1.66 $\times$
fmmPotential	F	12/16	25%	0.0979 $\pm$ 0.00128	0.0981 $\pm$ 0.00004	0.0957$\pm$0.00247	1.02 $\times$	1.00 $\times$	1.02 $\times$
sumEnergy	F	12/16	25%	0.0383 $\pm$ 0.00093	0.0549 $\pm$ 0.00122	0.0321$\pm$0.00002	1.20 $\times$	1.43 $\times$	1.71 $\times$
sumMass	F	10/16	38%	0.0372 $\pm$ 0.00002	0.0447 $\pm$ 0.00007	0.0274$\pm$0.00002	1.36 $\times$	1.20 $\times$	1.63 $\times$
clearFlags	M	15/16	6%	0.1534 $\pm$ 0.00235	0.1518$\pm$0.00161	0.1765 $\pm$ 0.00145	0.87 $\times$	0.99 $\times$	0.86 $\times$
scaleEnergy	M	16/16	0%	0.1508$\pm$0.00123	0.1520 $\pm$ 0.00212	0.1773 $\pm$ 0.00113	0.85 $\times$	1.01 $\times$	0.86 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0453 $\pm$ 0.00004	0.0561 $\pm$ 0.00002	0.0336$\pm$0.00016	1.35 $\times$	1.24 $\times$	1.67 $\times$
quantizationErrorProxy	F	10/25	60%	0.0491 $\pm$ 0.00001	0.0670 $\pm$ 0.00001	0.0433$\pm$0.00002	1.13 $\times$	1.36 $\times$	1.55 $\times$
Total				0.6896	0.7757	0.6823	1.01 $\times$	1.12 $\times$	1.14 $\times$
Geomean						1.13 $\times$	1.20 $\times$	1.36 $\times$	

Table 7: Per-pass PAPI counters for **OctTree** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
barnesHutPotential	F	2.2e+08/ 2.0e+08	8.2e+08 /1.0e+09	7.1e+04 /6.0e+05	1.1e+02/ 7.6e+01	2.0e+03 /2.3e+04	1.1e+02/ 7.6e+01	6.3e+06/ 2.4e+06
countActive	F	1.9e+08/ 1.5e+08	5.8e+08/ 5.8e+08	8.8e+05/ 4.0e+05	2.0e+02/ 5.8e+01	4.2e+03 /1.4e+04	2.0e+02/ 6.5e+01	6.1e+06/ 3.4e+05
countParticles	F	1.9e+08/ 1.4e+08	5.7e+08/ 5.7e+08	7.9e+05/ 2.1e+05	3.1e+02/ 5.2e+01	3.9e+03 /7.6e+03	3.1e+02/ 5.6e+01	5.9e+06/ 6.9e+04
fmmPotential	F	5.0e+08/ 4.9e+08	1.6e+09 /1.8e+09	2.4e+04 /7.9e+05	4.5e+02/ 3.2e+02	1.7e+03 /2.6e+04	4.6e+02/ 3.2e+02	4.4e+06/ 2.2e+06
sumEnergy	F	1.9e+08/ 1.6e+08	7.7e+08 /8.9e+08	2.4e+05 /6.4e+05	3.1e+02/ 5.9e+01	3.4e+03 /2.3e+04	3.1e+02/ 6.0e+01	6.2e+06/ 2.8e+06
sumMass	F	1.9e+08/ 1.4e+08	5.7e+08 /6.3e+08	6.7e+05/ 1.5e+05	1.4e+02/ 6.2e+01	3.0e+03 /8.2e+03	1.4e+02/ 6.3e+01	5.9e+06/ 1.3e+06
clearFlags	M	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	8.6e+05 /2.5e+06	1.6e+05 /4.3e+05	9.4e+03 /3.7e+04	4.7e+03 /1.2e+04	5.3e+06/ 2.5e+06
scaleEnergy	M	3.6e+08 /4.4e+08	9.1e+08 /1.9e+09	1.0e+06 /2.9e+06	2.4e+05 /4.1e+05	9.8e+03 /4.2e+04	6.3e+03 /1.4e+04	5.3e+06/ 2.7e+06
paletteEntriesQuantized	F	2.3e+08/ 1.7e+08	6.7e+08 /7.6e+08	5.5e+06/ 1.5e+06	2.6e+02/ 1.4e+02	1.2e+04 /4.2e+04	2.6e+02/ 1.4e+02	9.3e+06/ 1.4e+06
quantizationErrorProxy	F	2.5e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.3e+06/ 1.1e+06	5.5e+01 /2.7e+02	1.6e+04 /3.4e+04	6.0e+01 /2.7e+02	9.2e+06/ 3.2e+06
Total		2.7e+09/2.5e+09	8.2e+09/1.1e+10	1.4e+07/1.1e+07	4.1e+05/8.4e+05	6.6e+04/2.6e+05	1.3e+04/2.7e+04	6.4e+07/1.9e+07

Table 8: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
blockCountPass	F	2/9	78%	0.0131 $\pm$ 0.00001	0.0822 $\pm$ 0.00097	0.0019$\pm$0.00000	6.82 $\times$	6.28 $\times$	42.81 $\times$
branchStatsPass	F	2/9	78%	0.0127 $\pm$ 0.00052	0.0821 $\pm$ 0.00091	0.0036$\pm$0.00000	3.53 $\times$	6.48 $\times$	22.90 $\times$
castInstCountPass	F	3/9	67%	0.0195 $\pm$ 0.00002	0.0824 $\pm$ 0.00009	0.0013$\pm$0.00000	15.14 $\times$	4.22 $\times$	63.86 $\times$
has cycle	F	4/9	56%	0.0252 $\pm$ 0.00014	0.0943 $\pm$ 0.00187	0.0178$\pm$0.00015	1.42 $\times$	3.75 $\times$	5.31 $\times$
instCountPass	F	2/9	78%	0.0131 $\pm$ 0.00002	0.0839 $\pm$ 0.00009	0.0023$\pm$0.00001	5.77 $\times$	6.38 $\times$	36.86 $\times$
latencyModelPass	F	3/9	67%	0.0129 $\pm$ 0.00001	0.0821 $\pm$ 0.00086	0.0030$\pm$0.00000	4.33 $\times$	6.38 $\times$	27.62 $\times$
memoryOpStatsPass	F	3/9	67%	0.0128 $\pm$ 0.00001	0.0831 $\pm$ 0.00013	0.0038$\pm$0.00001	3.33 $\times$	6.51 $\times$	21.66 $\times$
throughputModelPass	F	3/9	67%	0.0132 $\pm$ 0.00004	0.0821 $\pm$ 0.00009	0.0030$\pm$0.00000	4.40 $\times$	6.23 $\times$	27.41 $\times$
verifyIR	F	9/9	0%	0.0134 $\pm$ 0.00006	0.0725 $\pm$ 0.00152	0.0034$\pm$0.00002	3.88 $\times$	5.42 $\times$	21.02 $\times$
stripSideEffectsPass	M	7/9	22%	0.0807$\pm$0.00019	0.1892 $\pm$ 0.00099	0.0901 $\pm$ 0.00014	0.90 $\times$	2.34 $\times$	2.10 $\times$
targetReturnPass	M	9/9	0%	0.0805$\pm$0.00018	0.1887 $\pm$ 0.00083	0.0913 $\pm$ 0.00267	0.88 $\times$	2.34 $\times$	2.07 $\times$
Total				0.2970	1.123	0.2215	1.34 $\times$	3.78 $\times$	5.07 $\times$
Geomean						3.34 $\times$	4.81 $\times$	16.06 $\times$	

Table 9: Per-pass PAPI counters for **Compiler** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
blockCountPass	F	6.6e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.3e+06/ 6.4e+03	2.7e+02/ 2.3e+01	2.4e+05/ 1.1e+03	2.7e+02/ 2.3e+01	4.6e+06/ 5.9e+02
branchStatsPass	F	6.4e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.4e+05/ 2.1e+04	2.4e+02/ 4.4e+01	5.7e+04/ 3.2e+03	2.4e+02/ 4.6e+01	4.6e+06/ 4.4e+05
castInstCountPass	F	9.8e+07/ 6.5e+06	5.1e+07/ 3.9e+07	1.3e+04 /2.2e+04	9.1e+01/ 1.8e+01	8.8e+02 /1.2e+03	9.2e+01/ 1.7e+01	4.6e+06/ 4.0e+03
has cycle	F	1.3e+08/ 9.0e+07	1.1e+08 /1.5e+08	2.4e+06/ 2.7e+05	5.0e+02/ 1.3e+02	1.5e+05/ 2.2e+04	5.1e+02/ 1.4e+02	5.1e+06/ 2.3e+06
instCountPass	F	6.6e+07/ 1.1e+07	5.6e+07 /5.7e+07	1.3e+06/ 1.3e+04	3.5e+02/ 3.1e+01	2.3e+05/ 1.1e+03	3.5e+02/ 3.2e+01	4.7e+06/ 3.5e+03
latencyModelPass	F	6.5e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	3.2e+02/ 3.6e+01	2.1e+05/ 3.7e+03	3.2e+02/ 3.6e+01	4.7e+06/ 5.3e+05
memoryOpStatsPass	F	6.4e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.9e+05/ 2.9e+04	2.1e+02/ 4.1e+01	3.7e+04/ 3.2e+03	2.1e+02/ 4.2e+01	4.6e+06/ 4.5e+05
throughputModelPass	F	6.5e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	4.6e+02/ 3.9e+01	2.1e+05/ 3.5e+03	4.6e+02/ 4.2e+01	4.8e+06/ 5.5e+05
verifyIR	F	6.7e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.3e+05/ 1.6e+04	3.5e+02/ 1.1e+02	1.5e+05/ 4.0e+03	3.4e+02/ 1.1e+02	4.6e+06/ 4.6e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.7e+06/ 1.3e+06	1.1e+05 /1.4e+05	6.7e+03 /1.9e+04	5.7e+03 /1.7e+04	4.0e+06/ 1.3e+06
targetReturnPass	M	1.5e+08 /2.0e+08	1.7e+08 /6.3e+08	1.7e+06/ 1.6e+06	1.0e+05 /2.0e+05	6.4e+03 /2.3e+04	5.6e+03 /2.3e+04	4.1e+06/ 1.5e+06
Total		9.9e+08/6.0e+08	9.9e+08/1.9e+09	1.3e+07/3.3e+06	2.2e+05/3.4e+05	1.3e+06/8.5e+04	1.4e+04/4.1e+04	5.0e+07/7.5e+06

Table 10: Per-pass performance for DBQuery, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countJoins	F	3/15	80%	0.0196 $\pm$ 0.00001	0.0288 $\pm$ 0.00001	0.0110$\pm$0.00001	1.79$\times$	1.47$\times$	2.63$\times$
filterSelectivitySkew	F	5/15	67%	0.0196 $\pm$ 0.00069	0.0289 $\pm$ 0.00001	0.0124$\pm$0.00002	1.58$\times$	1.47$\times$	2.33$\times$
hashJoinPressure	F	5/15	67%	0.0203 $\pm$ 0.00001	0.0330 $\pm$ 0.00001	0.0125$\pm$0.00001	1.63$\times$	1.63$\times$	2.64$\times$
sumCost	F	6/15	60%	0.0196 $\pm$ 0.00001	0.0287 $\pm$ 0.00013	0.0141$\pm$0.00091	1.39$\times$	1.46$\times$	2.03$\times$
sumMemory	F	6/15	60%	0.0197 $\pm$ 0.00001	0.0288 $\pm$ 0.00001	0.0144$\pm$0.00002	1.37$\times$	1.46$\times$	2.00$\times$
sumRows	F	6/15	60%	0.0269 $\pm$ 0.00001	0.0305 $\pm$ 0.00103	0.0218$\pm$0.00002	1.23$\times$	1.13$\times$	1.40$\times$
clearQueryFlags	M	14/15	7%	0.0689$\pm$0.00012	0.0763 $\pm$ 0.00127	0.0844 $\pm$ 0.00129	0.82 $\times$	1.11$\times$	0.90 $\times$
scaleCosts	M	15/15	0%	0.0684$\pm$0.00012	0.0765 $\pm$ 0.00010	0.0836 $\pm$ 0.00203	0.82 $\times$	1.12$\times$	0.91 $\times$
Total				0.2631	0.3313	0.2542	1.04 $\times$	1.26$\times$	1.30$\times$
Geomean						1.28$\times$	1.34$\times$	1.72$\times$	

Table 11: Per-pass PAPI counters for DBQuery (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countJoins	F	9.9e+07/ 5.5e+07	7.4e+07/ 7.1e+07	7.0e+05/ 2.6e+04	3.8e+02/ 7.6e+01	1.7e+05/ 2.1e+03	3.8e+02/ 6.4e+01	2.7e+06/ 2.2e+03
filterSelectivitySkew	F	9.9e+07/ 6.3e+07	1.0e+08 /1.1e+08	5.2e+05/ 7.3e+04	2.6e+02/ 5.3e+01	1.2e+05/ 6.2e+03	2.6e+02/ 5.3e+01	2.7e+06/ 2.5e+05
hashJoinPressure	F	1.0e+08/ 6.3e+07	8.4e+07 /9.9e+07	7.0e+05/ 9.5e+04	3.2e+02/ 9.0e+01	1.8e+05/ 7.2e+03	3.2e+02/ 9.4e+01	2.7e+06/ 1.4e+05
sumCost	F	9.8e+07/ 7.1e+07	9.0e+07 /1.2e+08	6.7e+05/ 8.3e+04	2.4e+02/ 6.5e+01	1.5e+05/ 1.1e+04	2.4e+02/ 7.2e+01	2.7e+06/ 3.5e+05
sumMemory	F	9.9e+07/ 7.3e+07	9.0e+07 /1.2e+08	5.7e+05/ 9.0e+04	3.9e+02/ 6.9e+01	1.3e+05/ 1.1e+04	3.9e+02/ 7.0e+01	2.7e+06/ 3.4e+05
sumRows	F	1.4e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.6e+05/ 1.1e+05	2.1e+02/ 8.0e+01	2.2e+04/ 7.5e+03	2.1e+02/ 8.1e+01	2.3e+06/ 2.9e+05
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.5e+05 /1.3e+06	7.6e+04 /2.6e+05	2.8e+04/ 2.4e+04	2.3e+03 /1.4e+04	2.4e+06/ 9.3e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.5e+05 /1.2e+06	7.4e+04 /1.9e+05	2.9e+04/ 2.9e+04	2.5e+03 /1.7e+04	2.4e+06/ 1.1e+06
Total		9.6e+08/8.8e+08	1.1e+09/2.2e+09	3.9e+06/3.0e+06	1.5e+05/4.5e+05	8.2e+05/9.7e+04	6.7e+03/3.2e+04	2.1e+07/3.4e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
classify Batch	F	5/7	29%	3.383 $\pm$ 0.0057	4.406 $\pm$ 0.0066	2.609$\pm$0.0048	1.30$\times$	1.30$\times$	1.69$\times$
classify Depth	F	4/7	43%	0.0487 $\pm$ 0.00001	0.0630 $\pm$ 0.00001	0.0368$\pm$0.00006	1.32$\times$	1.29$\times$	1.71$\times$
classify tree	F	5/7	29%	0.0483 $\pm$ 0.00082	0.0630 $\pm$ 0.00001	0.0367$\pm$0.00101	1.31$\times$	1.31$\times$	1.71$\times$
countFeatureUses	F	3/7	57%	0.0516 $\pm$ 0.00085	0.1419 $\pm$ 0.00186	0.0326$\pm$0.00002	1.58$\times$	2.75$\times$	4.35$\times$
countLeaves	F	2/7	71%	0.0490 $\pm$ 0.00073	0.1410 $\pm$ 0.00002	0.0251$\pm$0.00001	1.95$\times$	2.87$\times$	5.61$\times$
countNodes	F	2/7	71%	0.0488 $\pm$ 0.00112	0.1408 $\pm$ 0.00009	0.0262$\pm$0.00003	1.86$\times$	2.89$\times$	5.37$\times$
countSmallLeaves	F	3/7	57%	0.0526 $\pm$ 0.00003	0.1369 $\pm$ 0.00003	0.0325$\pm$0.00003	1.62$\times$	2.60$\times$	4.22$\times$
inferenceCost	F	2/7	71%	0.0519 $\pm$ 0.00170	0.1400 $\pm$ 0.00117	0.0289$\pm$0.00001	1.80$\times$	2.70$\times$	4.85$\times$
sumImpurity	F	3/7	57%	0.0517 $\pm$ 0.00002	0.1399 $\pm$ 0.00026	0.0324$\pm$0.00003	1.59$\times$	2.70$\times$	4.31$\times$
sumPathLengths	F	3/7	57%	0.0534 $\pm$ 0.00002	0.1394 $\pm$ 0.00085	0.0345$\pm$0.00001	1.55$\times$	2.61$\times$	4.05$\times$
sumSamples	F	3/7	57%	0.0497 $\pm$ 0.00002	0.1351 $\pm$ 0.00128	0.0326$\pm$0.00002	1.53$\times$	2.72$\times$	4.15$\times$
treeDepth	F	2/7	71%	0.0524 $\pm$ 0.00002	0.1399 $\pm$ 0.00243	0.0282$\pm$0.00100	1.86$\times$	2.67$\times$	4.96$\times$
Total				3.941	5.787	2.955	1.33$\times$	1.47$\times$	1.96$\times$
Geomean						1.59$\times$	2.26$\times$	3.60$\times$	

Table 13: Per-pass PAPI counters for DecisionTree (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
classify Batch	F	1.7e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	1.3e+05/ 3.7e+04	1.7e+06 /2.0e+06	1.3e+05/ 3.5e+04	9.3e+08/ 1.8e+07
classify Depth	F	2.5e+08/ 1.9e+08	4.7e+08 /8.5e+08	6.7e+06/ 3.7e+05	1.9e+03/ 8.1e+02	1.5e+05/ 2.8e+04	1.9e+03/ 7.5e+02	1.3e+07/ 2.8e+05
classify tree	F	2.4e+08/ 1.9e+08	4.7e+08 /8.5e+08	6.6e+06/ 3.7e+05	1.8e+03/ 4.4e+02	2.1e+04 /2.8e+04	1.8e+03/ 4.2e+02	1.3e+07/ 2.9e+05
countFeatureUses	F	2.6e+08/ 1.7e+08	7.5e+08 /7.8e+08	7.5e+06/ 1.0e+06	1.3e+03/ 8.8e+02	3.3e+04/ 1.4e+04	1.3e+03/ 8.8e+02	1.5e+07/ 2.0e+06
countLeaves	F	2.5e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.3e+06/ 1.1e+05	1.5e+03/ 4.1e+02	5.2e+04/ 8.1e+03	1.4e+03/ 4.0e+02	1.4e+07/ 3.5e+05
countNodes	F	2.5e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.3e+06/ 1.1e+05	6.5e+02/ 3.2e+02	5.1e+04/ 8.1e+03	6.5e+02/ 3.2e+02	1.4e+07/ 3.7e+05
countSmallLeaves	F	2.7e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 1.9e+05	7.9e+02/ 7.1e+02	2.9e+04/ 1.3e+04	7.9e+02/ 7.1e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.6e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.0e+06/ 1.2e+05	1.8e+03/ 4.0e+02	1.8e+04/ 8.0e+03	1.8e+03/ 4.0e+02	1.4e+07/ 3.5e+05
sumImpurity	F	2.6e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.6e+06/ 1.1e+06	6.5e+02 /7.0e+02	3.6e+04/ 1.3e+04	6.5e+02 /7.0e+02	1.4e+07/ 2.0e+06
sumPathLengths	F	2.7e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.9e+06/ 1.2e+06	1.1e+03/ 9.6e+02	4.1e+04/ 1.4e+04	1.1e+03/ 9.6e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.5e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.3e+06/ 4.5e+05	9.0e+02/ 8.6e+02	4.1e+04/ 1.4e+04	9.0e+02/ 8.6e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.7e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.1e+05	6.5e+02/ 4.0e+02	1.9e+04/ 8.2e+03	6.5e+02/ 3.9e+02	1.4e+07/ 3.4e+05
Total		2.0e+10/1.5e+10	4.0e+10/6.8e+10	5.4e+08/3.2e+07	1.4e+05/4.4e+04	2.2e+06/2.1e+06	1.4e+05/4.2e+04	1.1e+09/3.0e+07

Table 14: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
count styled	F	3/15	80%	0.0368 $\pm$ 0.00001	0.0718 $\pm$ 0.00003	0.0117$\pm$0.00001	3.15$\times$	1.95$\times$	6.16$\times$
find max Bottom	F	5/15	67%	0.0369 $\pm$ 0.00002	0.0714 $\pm$ 0.00101	0.0191$\pm$0.00003	1.94$\times$	1.93$\times$	3.75$\times$
SumArea	F	6/15	60%	0.0382 $\pm$ 0.00002	0.0723 $\pm$ 0.00008	0.0194$\pm$0.00003	1.97$\times$	1.90$\times$	3.74$\times$
sumTextWidth	F	3/15	80%	0.0366 $\pm$ 0.00001	0.0724 $\pm$ 0.00001	0.0117$\pm$0.00001	3.12$\times$	1.98$\times$	6.17$\times$
computeWidths	M	14/15	7%	0.0299$\pm$0.00089	0.0305 $\pm$ 0.00005	0.0436 $\pm$ 0.00006	0.68 $\times$	1.02 $\times$	0.70 $\times$
scaleLayout	M	15/15	0%	0.0287$\pm$0.00003	0.0302 $\pm$ 0.00004	0.0341 $\pm$ 0.00004	0.84 $\times$	1.05 $\times$	0.88 $\times$
Total				0.2069	0.3487	0.1396	1.48$\times$	1.68$\times$	2.50$\times$
Geomean						1.67$\times$	1.57$\times$	2.63$\times$	

Table 15: Per-pass PAPI counters for DomTree (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
count styled	F	1.9e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.3e+06/ 2.4e+04	6.8e+02/ 9.2e+01	3.1e+05/ 4.8e+03	6.8e+02/ 9.4e+01	1.4e+07/ 6.5e+05
find max Bottom	F	1.9e+08/ 9.6e+07	2.5e+08 /4.1e+08	2.7e+06/ 1.9e+06	5.5e+02/ 5.6e+01	1.3e+05/ 1.0e+04	5.6e+02/ 6.0e+01	1.4e+07/ 1.4e+06
SumArea	F	1.9e+08/ 9.8e+07	2.9e+08 /4.9e+08	2.9e+06/ 1.2e+06	6.9e+02/ 5.5e+01	1.8e+05/ 1.2e+04	6.9e+02/ 5.5e+01	1.3e+07/ 1.9e+06
sumTextWidth	F	1.9e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.5e+06/ 2.2e+04	7.2e+02/ 8.0e+01	4.1e+05/ 4.2e+03	7.2e+02/ 8.1e+01	1.3e+07/ 6.6e+05
computeWidths	M	6.5e+07 /1.3e+08	1.4e+08 /4.6e+08	1.2e+06 /1.4e+06	5.4e+04/ 4.5e+04	6.7e+03 /1.2e+04	2.0e+03 /1.5e+04	1.2e+06/ 4.4e+05
scaleLayout	M	6.0e+07 /8.6e+07	8.8e+07 /2.9e+08	1.0e+06/ 7.0e+05	5.2e+04 /6.3e+04	1.6e+04/ 8.9e+03	3.9e+03 /1.4e+04	1.3e+06/ 4.6e+05
Total		8.8e+08/5.3e+08	1.2e+09/2.2e+09	1.3e+07/5.2e+06	1.1e+05/1.1e+05	1.1e+06/5.2e+04	8.6e+03/2.9e+04	5.6e+07/5.5e+06

Table 16: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countInRange tight_box	F	13/17	24%	0.0296 $\pm$ 0.00013	0.0421 $\pm$ 0.00100	0.0201$\pm$0.00001	1.48$\times$	1.42$\times$	2.10$\times$
Find nearest Neighbour	F	13/17	24%	0.0294 $\pm$ 0.00096	0.0414 $\pm$ 0.00066	0.0212$\pm$0.00003	1.38$\times$	1.41$\times$	1.95$\times$
photonMappingBenchmark	F	12/17	29%	0.0440 $\pm$ 0.00002	0.0495 $\pm$ 0.00002	0.0411$\pm$0.00001	1.07 $\times$	1.12$\times$	1.21$\times$
pointCloudNeighborhood	F	11/17	35%	0.0264 $\pm$ 0.00001	0.0406 $\pm$ 0.00002	0.0167$\pm$0.00001	1.58$\times$	1.54$\times$	2.43$\times$
sumMassInRange	F	14/17	18%	0.0289 $\pm$ 0.00001	0.0422 $\pm$ 0.00001	0.0200$\pm$0.00001	1.44$\times$	1.46$\times$	2.10$\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0306 $\pm$ 0.00001	0.0412 $\pm$ 0.00002	0.0209$\pm$0.00000	1.46$\times$	1.34$\times$	1.97$\times$
Total				0.1890	0.2570	0.1401	1.35$\times$	1.36$\times$	1.83$\times$
Geomean						1.39$\times$	1.38$\times$	1.92$\times$	

Table 17: Per-pass PAPI counters for **KDTree** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countInRange tight_box	F	1.5e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.2e+06/ 1.7e+06	1.9e+02 /2.0e+02	3.3e+03 /1.9e+04	1.9e+02 /2.0e+02	7.7e+06/ 1.9e+06
Find nearest Neighbour	F	1.5e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.8e+06/ 1.7e+06	9.6e+01/ 7.3e+01	4.3e+03 /1.8e+04	9.6e+01/ 7.5e+01	7.7e+06/ 1.8e+06
photonMappingBenchmark	F	2.2e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	2.4e+02 /4.9e+02	3.5e+03 /1.6e+04	2.4e+02 /4.9e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.3e+08/ 8.5e+07	3.3e+08 /4.8e+08	2.7e+06/ 8.8e+05	2.1e+02/ 1.3e+02	5.0e+03 /1.7e+04	2.2e+02/ 1.3e+02	7.7e+06/ 1.6e+06
sumMassInRange	F	1.5e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.6e+06/ 2.4e+06	8.1e+01 /1.6e+02	4.0e+03 /2.1e+04	8.2e+01 /1.6e+02	7.7e+06/ 2.2e+06
twoPointCorrelation bin_8_16	F	1.6e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.1e+06/ 7.7e+05	1.3e+02/ 7.3e+01	3.0e+03 /1.4e+04	1.3e+02/ 7.2e+01	7.6e+06/ 1.3e+06
Total		9.6e+08/7.1e+08	3.1e+09/4.0e+09	1.4e+07/9.9e+06	9.4e+02/1.1e+03	2.3e+04/1.0e+05	9.5e+02/1.1e+03	4.4e+07/1.0e+07

Table 18: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
reduction	F	2/11	82%	0.0319 $\pm$ 0.00001	0.1399 $\pm$ 0.00006	0.0050$\pm$0.00002	6.35$\times$	4.38$\times$	27.83$\times$
Total				0.0319	0.1399	0.0050	6.35$\times$	4.38$\times$	27.83$\times$
Geomean						6.35$\times$	4.38$\times$	27.83$\times$	

Table 19: Per-pass PAPI counters for **LinearListReduction** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
reduction	F	1.6e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.6e+06/ 4.6e+04	2.7e+02/ 4.7e+01	2.5e+05/ 7.1e+03	2.7e+02/ 4.8e+01	1.2e+07/ 1.1e+06
Total		1.6e+08/2.5e+07	9.0e+07/1.0e+08	1.6e+06/4.6e+04	2.7e+02/4.7e+01	2.5e+05/7.1e+03	2.7e+02/4.8e+01	1.2e+07/1.1e+06

Table 20: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
length List	F	1/2	50%	0.0429 $\pm$ 0.00071	<i>OOM</i>	0.0217$\pm$0.00007	1.98$\times$	–	–
sumList	F	2/2	0%	0.0451$\pm$0.00002	<i>OOM</i>	0.0471 $\pm$ 0.00040	0.96 $\times$	–	–
sumList tail recursive	F	2/2	0%	0.0454$\pm$0.00002	<i>OOM</i>	0.0466 $\pm$ 0.00002	0.97 $\times$	–	–
add1 List	M	2/2	0%	0.2542$\pm$0.00117	<i>OOM</i>	0.3424 $\pm$ 0.00333	0.74 $\times$	–	–
Total				0.3875	–	0.4577	0.85 $\times$	–	–
Geomean						1.08 $\times$	–	–	

Table 21: Per-pass PAPI counters for **List** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
length List	F	2.2e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 8.8e+04	1.3e+02/ 6.8e+01	3.0e+03 /1.4e+04	1.4e+02/ 6.8e+01	1.3e+07/ 1.1e+06
sumList	F	2.3e+08 /2.4e+08	9.0e+08 /1.0e+09	6.7e+05/ 5.5e+05	1.1e+02/ 9.3e+01	3.3e+03 /6.9e+04	1.1e+02/ 9.5e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.3e+08 /2.4e+08	9.0e+08 /1.0e+09	5.2e+05 /5.8e+05	1.4e+02 /1.8e+02	3.1e+03 /5.9e+04	1.4e+02 /1.8e+02	1.3e+07/ 1.2e+07
add1 List	M	5.9e+08 /9.7e+08	1.9e+09 /2.8e+09	7.8e+06/ 1.3e+06	2.9e+05 /6.1e+05	1.2e+04 /5.6e+04	1.2e+04 /1.8e+04	9.2e+06/ 7.7e+06
Total		1.3e+09/1.6e+09	4.5e+09/5.6e+09	9.2e+06/2.6e+06	2.9e+05/6.1e+05	2.2e+04/2.0e+05	1.3e+04/1.8e+04	4.9e+07/3.3e+07

Table 22: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sumTree	F	3/3	0%	0.0091$\pm$0.00000	0.0406 $\pm$ 0.00002	0.0116 $\pm$ 0.00001	0.79 $\times$	4.44$\times$	3.51$\times$
sumTree TailRec	F	3/3	0%	0.0086$\pm$0.00001	0.0402 $\pm$ 0.00001	0.0113 $\pm$ 0.00002	0.76 $\times$	4.65$\times$	3.55$\times$
add1Tree	M	3/3	0%	0.0459 $\pm$ 0.00089	0.0607 $\pm$ 0.00122	0.0432$\pm$0.00051	1.06 $\times$	1.32$\times$	1.41$\times$
Total				0.0636	0.1415	0.0661	0.96 $\times$	2.22$\times$	2.14$\times$
Geomean						0.86 $\times$	3.01$\times$	2.60$\times$	

Table 23: Per-pass PAPI counters for **MonoTree** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
sumTree	F	4.6e+07 /5.8e+07	2.2e+08 /2.6e+08	7.3e+03 /3.7e+04	2.4e+02/ 2.0e+02	7.3e+02 /4.0e+03	2.4e+02/ 2.0e+02	7.7e+05/ 6.2e+05
sumTree TailRec	F	4.4e+07 /5.7e+07	1.9e+08 /2.2e+08	6.6e+03 /2.5e+04	2.9e+02/ 2.1e+02	8.0e+02 /4.1e+03	3.0e+02/ 2.1e+02	7.6e+05/ 6.1e+05
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.1e+05 /3.8e+05	4.9e+04/ 4.6e+04	2.1e+03 /5.2e+03	1.6e+03 /1.6e+03	5.1e+05/ 4.5e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.2e+05/4.5e+05	5.0e+04/4.6e+04	3.6e+03/1.3e+04	2.1e+03/2.1e+03	2.0e+06/1.7e+06

Table 24: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countLargeItems	F	3/5	40%	0.0138 $\pm$ 0.00001	0.0447 $\pm$ 0.00001	0.0110$\pm$0.00002	1.26$\times$	3.24$\times$	4.07$\times$
countMarked	F	3/5	40%	0.0137 $\pm$ 0.00000	0.0447 $\pm$ 0.00012	0.0109$\pm$0.00002	1.25$\times$	3.27$\times$	4.09$\times$
countSurvivors	F	4/5	20%	0.0146 $\pm$ 0.00001	0.0448 $\pm$ 0.00002	0.0135$\pm$0.00001	1.07 $\times$	3.08$\times$	3.30$\times$
deadBytes	F	4/5	20%	0.0143 $\pm$ 0.00001	0.0441 $\pm$ 0.00118	0.0128$\pm$0.00001	1.12$\times$	3.08$\times$	3.45$\times$
liveBytes	F	4/5	20%	0.0147 $\pm$ 0.00001	0.0435 $\pm$ 0.00001	0.0128$\pm$0.00001	1.15$\times$	2.96$\times$	3.40$\times$
sumObjIds	F	3/5	40%	0.0133 $\pm$ 0.00000	0.0433 $\pm$ 0.00001	0.0106$\pm$0.00007	1.25$\times$	3.24$\times$	4.06$\times$
totalHeapSize	F	3/5	40%	0.0133 $\pm$ 0.00001	0.0433 $\pm$ 0.00024	0.0104$\pm$0.00091	1.28$\times$	3.24$\times$	4.16$\times$
sweepUnmarked	M	5/5	0%	0.0718$\pm$0.00022	0.0878 $\pm$ 0.00114	0.0804 $\pm$ 0.00125	0.89 $\times$	1.22$\times$	1.09 $\times$
touchHotObjects	M	5/5	0%	0.0709$\pm$0.00006	0.0872 $\pm$ 0.00007	0.0812 $\pm$ 0.00018	0.87 $\times$	1.23$\times$	1.07 $\times$
Total				0.2403	0.4832	0.2437	0.99 $\times$	2.01$\times$	1.98$\times$
Geomean						1.12$\times$	2.56$\times$	2.86$\times$	

Table 25: Per-pass PAPI counters for `ObjectGraph` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countLargeItems	F	7.0e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.1e+06/ 4.2e+04	1.6e+02/ 1.5e+02	3.2e+03 /3.6e+03	1.6e+02/ 1.5e+02	2.8e+06/ 6.4e+05
countMarked	F	6.9e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.4e+04	1.6e+02/ 1.2e+02	3.4e+03 /3.9e+03	1.6e+02/ 1.2e+02	2.9e+06/ 6.5e+05
countSurvivors	F	7.4e+07/ 6.9e+07	3.1e+08 /3.8e+08	1.0e+06/ 7.5e+04	2.0e+02 /2.4e+02	2.6e+03 /6.2e+03	2.1e+02 /2.4e+02	2.8e+06/ 1.2e+06
deadBytes	F	7.2e+07/ 6.5e+07	2.7e+08 /3.5e+08	7.7e+05/ 6.5e+04	1.9e+02/ 1.3e+02	2.3e+03 /5.4e+03	2.0e+02/ 1.3e+02	2.7e+06/ 1.2e+06
liveBytes	F	7.4e+07/ 6.5e+07	2.7e+08 /3.5e+08	7.1e+05/ 6.4e+04	2.6e+02/ 1.9e+02	2.1e+03 /5.4e+03	2.6e+02/ 1.9e+02	2.8e+06/ 1.2e+06
sumObjIds	F	6.8e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.3e+06/ 7.8e+04	2.0e+02/ 1.6e+02	3.1e+03 /4.4e+03	2.0e+02/ 1.6e+02	2.8e+06/ 6.4e+05
totalHeapSize	F	6.7e+07/ 5.2e+07	2.5e+08 /2.7e+08	1.3e+06/ 1.1e+05	2.0e+02/ 1.2e+02	3.0e+03 /3.7e+03	2.1e+02/ 1.2e+02	2.8e+06/ 6.4e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	9.9e+05/ 4.9e+05	6.8e+04 /1.5e+05	4.6e+03 /1.1e+04	3.7e+03 /9.0e+03	2.0e+06/ 1.1e+06
touchHotObjects	M	1.8e+08 /2.3e+08	5.7e+08 /9.0e+08	7.9e+05/ 5.4e+05	1.4e+05/ 1.2e+05	4.9e+03 /1.1e+04	5.5e+03 /8.1e+03	2.0e+06/ 1.1e+06
Total		8.7e+08/8.8e+08	3.0e+09/4.0e+09	9.2e+06/1.5e+06	2.1e+05/2.7e+05	2.9e+04/5.4e+04	1.1e+04/1.8e+04	2.4e+07/8.4e+06

Table 26: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autorefineMaxLevel	F	4/8	50%	0.0219 $\pm$ 0.00001	0.0452 $\pm$ 0.00002	0.0214$\pm$0.00001	1.02 $\times$	2.07$\times$	2.11$\times$
compressMass	F	3/8	62%	0.0209 $\pm$ 0.00001	0.0456 $\pm$ 0.00086	0.0116$\pm$0.00001	1.79$\times$	2.18$\times$	3.91$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0229 $\pm$ 0.00002	0.0476 $\pm$ 0.00002	0.0189$\pm$0.00002	1.21$\times$	2.08$\times$	2.51$\times$
norm2Estimate	F	5/8	38%	0.0263 $\pm$ 0.00001	0.0551 $\pm$ 0.00139	0.0187$\pm$0.00005	1.41$\times$	2.09$\times$	2.95$\times$
pmapCutHistogram	F	4/8	50%	0.0334 $\pm$ 0.00086	0.0563 $\pm$ 0.00001	0.0128$\pm$0.00001	2.60$\times$	1.69$\times$	4.38$\times$
truncateTolViolations	F	3/8	62%	0.0213 $\pm$ 0.00002	0.0464 $\pm$ 0.00001	0.0117$\pm$0.00001	1.83$\times$	2.18$\times$	3.98$\times$
addConstPW	M	8/8	0%	0.1200$\pm$0.00020	0.1301 $\pm$ 0.00027	0.1338 $\pm$ 0.00175	0.90 $\times$	1.08 $\times$	0.97 $\times$
diffPW	M	8/8	0%	0.1193$\pm$0.00021	0.1327 $\pm$ 0.00100	0.1334 $\pm$ 0.00017	0.89 $\times$	1.11$\times$	0.99 $\times$
Total				0.3860	0.5590	0.3623	1.07 $\times$	1.45$\times$	1.54$\times$
Geomean						1.36$\times$	1.75$\times$	2.38$\times$	

Table 27: Per-pass PAPI counters for `PiecewiseFunctions` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
autorefineMaxLevel	F	1.1e+08/ 1.1e+08	2.5e+08 /3.7e+08	2.5e+06/ 2.9e+05	5.6e+02/ 8.0e+01	9.7e+03/ 6.5e+03	5.6e+02/ 7.9e+01	6.1e+06/ 8.9e+05
compressMass	F	1.1e+08/ 5.9e+07	2.5e+08 /2.8e+08	2.2e+06/ 2.9e+04	5.9e+02/ 2.7e+02	1.9e+04/ 4.8e+03	5.9e+02/ 2.8e+02	6.2e+06/ 6.2e+05
lbDeuxLoadProxy	F	1.2e+08/ 9.6e+07	3.3e+08 /4.9e+08	2.4e+06/ 5.4e+05	3.3e+02/ 1.2e+02	1.4e+04/ 8.8e+03	3.3e+02/ 1.2e+02	6.2e+06/ 1.6e+06
norm2Estimate	F	1.3e+08/ 9.5e+07	3.4e+08 /4.4e+08	1.4e+06/ 6.6e+05	2.6e+02/ 1.1e+02	1.1e+04/ 9.6e+03	2.6e+02/ 1.1e+02	6.1e+06/ 1.5e+06
pmapCutHistogram	F	1.7e+08/ 6.5e+07	2.8e+08 /3.7e+08	1.5e+06/ 7.0e+04	1.3e+03/ 3.0e+02	9.1e+04/ 7.4e+03	1.3e+03/ 3.1e+02	4.7e+06/ 1.3e+06
truncateTolViolations	F	1.1e+08/ 5.9e+07	2.5e+08 /2.9e+08	2.1e+06/ 4.1e+04	6.0e+02/ 1.9e+02	1.6e+04/ 4.8e+03	6.0e+02/ 2.0e+02	6.3e+06/ 6.2e+05
addConstPW	M	2.8e+08 /3.5e+08	5.7e+08 /1.2e+09	1.1e+06 /2.5e+06	2.9e+05/ 2.1e+05	1.0e+04 /2.7e+04	3.9e+03 /2.1e+04	4.4e+06/ 1.8e+06
diffPW	M	2.8e+08 /3.5e+08	5.9e+08 /1.3e+09	1.2e+06 /2.4e+06	2.7e+05/ 1.7e+05	9.6e+03 /2.6e+04	5.2e+03 /2.0e+04	4.5e+06/ 1.8e+06
Total		1.3e+09/1.2e+09	2.8e+09/4.7e+09	1.4e+07/6.5e+06	5.6e+05/3.8e+05	1.8e+05/9.5e+04	1.3e+04/4.2e+04	4.5e+07/1.0e+07

Table 28: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sum tree	F	5/5	0%	0.0286 $\pm$ 0.00140	0.0505 $\pm$ 0.00121	0.0261$\pm$0.00009	1.09 $\times$	1.76$\times$	1.93$\times$
add 1 tree	M	5/5	0%	0.0835 $\pm$ 0.00083	0.0939 $\pm$ 0.00025	0.0784$\pm$0.00122	1.07 $\times$	1.12$\times$	1.20$\times$
Total				0.1121	0.1444	0.1045	1.07 $\times$	1.29$\times$	1.38$\times$
Geomean						1.08 $\times$	1.41$\times$	1.52$\times$	

Table 29: Per-pass PAPI counters for `TernaryTree` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	5.3e+04 /3.7e+05	7.1e+01 /1.2e+02	8.8e+02 /9.3e+03	7.1e+01 /1.2e+02	1.8e+06/ 1.3e+06
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.3e+05 /1.2e+06	6.8e+04 /1.4e+05	3.5e+03 /1.1e+04	2.4e+03 /2.7e+03	1.5e+06/ 1.1e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	4.8e+05/1.6e+06	6.8e+04/1.4e+05	4.4e+03/2.0e+04	2.5e+03/2.8e+03	3.3e+06/2.5e+06

Table 30: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autocompleteTopKProxy	F	5/10	50%	0.0152 $\pm$ 0.00054	0.0285 $\pm$ 0.00001	0.0086$\pm$0.00004	1.76$\times$	1.88$\times$	3.31$\times$
countLazyNodes	F	4/10	60%	0.0137 $\pm$ 0.00049	0.0289 $\pm$ 0.00001	0.0067$\pm$0.00001	2.03$\times$	2.12$\times$	4.29$\times$
countTerminals	F	3/10	70%	0.0129 $\pm$ 0.00001	0.0268 $\pm$ 0.00001	0.0058$\pm$0.00000	2.21$\times$	2.07$\times$	4.58$\times$
sumPrefixFreq	F	3/10	70%	0.0131 $\pm$ 0.00001	0.0273 $\pm$ 0.00076	0.0057$\pm$0.00007	2.29$\times$	2.08$\times$	4.76$\times$
sumSubtreeHints	F	3/10	70%	0.0131 $\pm$ 0.00000	0.0272 $\pm$ 0.00001	0.0057$\pm$0.00001	2.28$\times$	2.08$\times$	4.74$\times$
decayTrieStats	M	10/10	0%	0.0762$\pm$0.00016	0.0805 $\pm$ 0.00015	0.0849 $\pm$ 0.00300	0.90 $\times$	1.06 $\times$	0.95 $\times$
resetTraversalState	M	10/10	0%	0.0763$\pm$0.00122	0.0815 $\pm$ 0.00200	0.0818 $\pm$ 0.00009	0.93 $\times$	1.07 $\times$	1.00 $\times$
Total				0.2205	0.3007	0.1994	1.11$\times$	1.36$\times$	1.51$\times$
Geomean						1.66$\times$	1.70$\times$	2.81$\times$	

Table 31: Per-pass PAPI counters for **Trie** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
autocompleteTopKProxy	F	7.7e+07/ 4.3e+07	1.3e+08 /2.1e+08	1.6e+06/ 1.8e+05	6.6e+01/ 3.6e+01	4.3e+04/ 4.7e+03	7.2e+01/ 3.6e+01	3.7e+06/ 8.1e+05
countLazyNodes	F	6.9e+07/ 3.4e+07	1.3e+08 /1.6e+08	1.4e+06/ 2.2e+04	8.0e+01/ 7.1e+01	1.6e+04/ 3.3e+03	8.0e+01/ 7.0e+01	4.0e+06/ 5.7e+05
countTerminals	F	6.5e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.5e+06/ 1.1e+04	1.1e+02/ 6.3e+01	2.9e+04/ 2.3e+03	1.1e+02/ 6.1e+01	3.9e+06/ 2.2e+05
sumPrefixFreq	F	6.6e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.4e+04	7.7e+01/ 5.6e+01	2.2e+04/ 2.5e+03	8.0e+01/ 5.6e+01	3.9e+06/ 2.1e+05
sumSubtreeHints	F	6.6e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.6e+06/ 1.4e+04	1.0e+02/ 5.3e+01	2.9e+04/ 2.5e+03	1.1e+02/ 5.3e+01	3.9e+06/ 2.1e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	8.6e+05 /1.3e+06	1.4e+05/ 1.2e+05	6.7e+03 /2.2e+04	1.6e+03 /2.9e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	9.6e+05/ 8.2e+05	1.3e+05 /1.6e+05	7.2e+03 /1.4e+04	1.9e+03 /2.0e+04	3.0e+06/ 9.2e+05
Total		6.7e+08/5.7e+08	1.2e+09/2.2e+09	9.4e+06/2.4e+06	2.7e+05/2.8e+05	1.5e+05/5.1e+04	3.9e+03/4.9e+04	2.5e+07/4.2e+06